# Project Assessment 4

## Project 5: XAI for Cross-Domain Selection Perturbative Hyper-Heuristics

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## 1 Introduction

Assessment 4 completes the work planned in Section 5 of Assessment 3. The baseline run on a single domain with a pooled surrogate established that the measurement apparatus works; the seven items listed in that conclusion identified what had to change before the results could carry scientific weight. This report documents the outcome of those changes.

The core research question remains: *what does a hyper-heuristic implicitly learn to respond to, and does this differ between selection strategies, acceptance strategies, and search topologies?* Assessment 3 could not answer it because the pooled surrogate confounded search topology with acceptance behaviour, the objective-stability issue in the log return statistic inflated one heuristic's apparent utility by a factor of roughly thirty, LIME had been applied at only two decision points, and the entire run used stub problem domains rather than the real CHeSC HyFlex benchmark. All four of those deficiencies are corrected here.

The six changes made for Assessment 4 are: (1) real CHeSC HyFlex jar files used throughout; (2) per-technique surrogates trained and SHAP profiles compared across them; (3) the experimental matrix expanded to four techniques across six domains on a single instance and seed as a stepping stone toward the full planned matrix; (4) multi-point budget scaling settled; (5) log return statistic stabilised and medians and counts reported alongside means; (6) LIME applied over a grid of fifty decision points spanning the full budget.

## 2 Implementation Changes from Assessment 3

### 2.1 Real HyFlex domains

Assessment 3 noted that all runs executed against stub problem domains because the CHeSC HyFlex jar files were not yet available. That limitation is resolved. The jar files `chesc-fixed-no-ps.jar` and `chesc-ps.jar` are present under `lib/hyflex/` and all 24 runs in this assessment report `stub: false` in their manifests. Objective values are now directly comparable to published CHeSC results.

### 2.2 Per-technique surrogates

Assessment 3 trained a single surrogate on the pooled output of all four techniques. Section 4.2 of Assessment 3 showed that the dominant SHAP attributions (population rank and diversity)

were artefacts of the pooled dataset: in a single-point run those features take sentinel value $-1$, so a tree model splits on them before it considers anything else. The surrogate was modelling topology membership, not acceptance behaviour.

The correction is to train one gradient-boosting surrogate per technique on that technique's decisions only. SHAP profiles are then compared across techniques. A pooled surrogate is retained as a secondary output for topology classification but is not used for the primary explanation.

### 2.3 Log return stabilisation

Assessment 3, Section 4.6, identified that the log return statistic is unstable when the denominator approaches zero, causing one heuristic to appear thirty times more useful than the next best. The fix is a symmetric clip applied after the log transform: $\log(o_{\text{after}}/o_{\text{before}})$ is clamped to $[-10, +10]$. Median and interquartile range are reported alongside the mean, and application counts are shown so that a large mean on few applications is distinguishable from a genuinely useful heuristic.

### 2.4 LIME grid over the full budget

Assessment 3 applied LIME at two hand-picked decision points. The phase transition hypothesis (exploratory acceptance early, selective acceptance late) rested on those two points alone. The new pipeline runs LIME at fifty evenly-spaced quantile positions across the budget, records every condition produced at each point, and aggregates condition frequency by progress band. Phase structure is validated against acceptance rate measured directly from the traces, which requires no model at all.

### 2.5 Multi-point budget

Assessment 3, Section 4.7, noted that under an equal iteration budget a population of ten receives roughly a tenth of the perturbations per member that a single-point run applies to its incumbent, making MP-CF structurally disadvantaged on final solution quality. The implementation supports a `budgetScaling: per_member` option that multiplies the base iteration budget by the population size before running, giving each member a comparable perturbation count. Runs in this assessment use the unscaled budget to maintain consistency with Assessment 3 and the planned full matrix; the scaling option is available and the comparison is discussed in Section 4.

### 2.6 Interactive experiment runner

A Bash script (`scripts/run_domain.sh`) provides an interactive menu for selecting domain, technique, budget and seed, running the full set, and optionally invoking the Python analysis pipeline on completion. The script was used to produce all runs reported here.

## 3 Experimental Protocol

### 3.1 Domains and techniques

All six CHeSC HyFlex domains were run: Boolean Satisfiability (SAT), 1-D Bin Packing (BinPacking), Permutation Flow Shop Scheduling (FlowShop), Travelling Salesman Problem (TSP), Vehicle Routing Problem (VRP), and Personnel Scheduling (PersonnelScheduling). The four techniques are the same as Assessment 3 and are listed in Table 1.

All runs: instance 0, seed 42, 100,000 iterations (iteration-based budget). This gives a single-seed, single-instance cross-domain snapshot. Statistical replication across seeds is planned for the full matrix but not yet executed; no significance claims are made in this report.

| Technique | Topology | Selection | Acceptance |
|---|---|---|---|
| SP-SR-LA | Single-point | Simple Random | Late Acceptance, $L = 50$ |
| SP-CF-LA | Single-point | Choice Function | Late Acceptance, $L = 50$ |
| SP-CF-AILTA | Single-point | Choice Function | AILTA |
| MP-CF | Multi-point (pop. 10) | Choice Function | Population replacement |

Table 1: The four techniques, unchanged from Assessment 3.

### 3.2 Dataset

The 24 runs produce approximately 2,500,000 recorded decisions. Each decision is one row in the feature matrix, labelled with the technique, domain, and outcome. Table 2 records wall-clock time per domain; PersonnelScheduling is an order of magnitude slower than the other domains and dominated the total run time.

| Domain | SP-CF-LA | SP-CF-AILTA | SP-SR-LA | MP-CF |
|---|---|---|---|---|
| SAT | 239 s | 227 s | 234 s | 242 s |
| BinPacking | 199 s | 205 s | 216 s | 713 s |
| FlowShop | 52 s | 73 s | 52 s | 57 s |
| TSP | 14 s | 13 s | 9 s | 21 s |
| VRP | 14 s | 26 s | 12 s | 23 s |
| PersonnelScheduling | 20,851 s | 22,398 s | 20,572 s | 28,042 s |

Table 2: Wall-clock time per run. Iteration count is 100,000 in all cases. The per-iteration overhead differs significantly between domains.

## 4 Results

### 4.1 Final objective values

Table 3 reports the final objective value at the end of each run. All domains are minimisation problems; lower values are better. The best result per domain is shown in bold.

| Domain | SP-CF-LA | SP-CF-AILTA | SP-SR-LA | MP-CF |
|---|---|---|---|---|
| SAT | 11.0 | **2.0** | 16.0 | 49.0 |
| BinPacking | 0.01227 | 0.01728 | **0.01090** | 0.01111 |
| FlowShop | 6350.0 | **6297.0** | 6346.0 | 6388.0 |
| TSP | 52798.9 | 49283.8 | 49737.3 | **49193.3** |
| VRP | 6165.5 | 5197.6 | 6147.4 | **5176.1** |
| PersonnelScheduling | 3298.0 | **3282.0** | 3324.0 | 3335.0 |

Table 3: Final objective values across all 24 runs. All domains are minimisation problems. Instance 0, seed 42, 100,000 iterations.

### 4.2 Surrogate faithfulness

One surrogate is trained per technique on that technique's decisions only. Faithfulness is measured by five-fold stratified cross-validation. Table 4 reports accuracy and ROC-AUC for each technique surrogate alongside the majority-class baseline for that technique's dataset.

| Technique | Accuracy | ROC-AUC | Majority baseline |
|---|---|---|---|
| SP-CF-LA | $89.34\% \pm 0.11\%$ | $96.77\% \pm 0.08\%$ | 71.54% |
| SP-CF-AILTA | $97.43\% \pm 0.09\%$ | $94.96\% \pm 0.22\%$ | 95.32% |
| SP-SR-LA | $71.74\% \pm 0.01\%$ | $51.75\% \pm 0.35\%$ | 71.76% |
| MP-CF | $99.53\% \pm 0.02\%$ | $96.25\% \pm 0.26\%$ | 99.38% |

Table 4: Per-technique surrogate faithfulness (five-fold stratified CV). All single-point technique surrogates drop the three population features (constant at sentinel $-1$), leaving 13 features. The SP-SR-LA ROC-AUC of 51.75% is effectively random and requires analysis (Section 4.4). The high majority-class baselines for SP-CF-AILTA and MP-CF reflect that both methods accept rarely: AILTA tightens its threshold aggressively, and multi-point replacement events are infrequent by construction.

### 4.3 Cross-technique SHAP comparison

Table 5 shows the mean absolute SHAP value for each feature in each per-technique surrogate, ranked by average importance across techniques. Features with zero importance in all four surrogates are omitted.

| Feature | SP-CF-LA | SP-CF-AILTA | SP-SR-LA | MP-CF |
|---|---|---|---|---|
| feat_improve_rate_50 | 1.476 | 1.305 | 1.281 | 0.109 |
| feat_gap_to_best | 0.317 | 10.859 | 0.509 | 48.081 |
| feat_mean_logret_50 | 0.686 | 0.438 | 0.035 | 0.167 |
| feat_progress | 0.208 | 0.256 | 0.234 | 0.249 |
| feat_iters_since_improve | 0.155 | 0.317 | 0.212 | 45.603 |
| feat_last_h1 | 0.389 | 0.275 | 0.061 | 0.016 |
| feat_last_h2 | 0.212 | 0.930 | 0.029 | 0.031 |
| feat_last_h3 | 0.433 | 0.737 | 0.077 | 0.062 |
| feat_obj_norm | 0.047 | 0.021 | 0.010 | 0.636 |
| feat_iters_since_accept | 0.000 | 0.000 | 0.000 | 86.231 |
| feat_llh_count | 1.740 | 11.219 | 1.695 | 1.137 |
| feat_last_class1 | 0.110 | 0.177 | 0.034 | 0.014 |
| feat_last_class2 | 0.102 | 0.020 | 0.042 | 0.007 |
| feat_last_class3 | 0.051 | 0.822 | 0.033 | 0.004 |
| feat_pop_diversity | — | — | — | 6.943 |
| feat_pop_target_rank | — | — | — | 0.361 |

Table 5: Mean absolute SHAP value per feature per technique. Population features (—) are dropped for all single-point techniques where they take sentinel value $-1$. MP-CF values are on a substantially different scale due to the extreme class imbalance (99.4% rejection rate); the ordinal ranking of features within each technique is more comparable than cross-technique magnitudes.

### 4.4 Log-return statistics

Table 6 reports per-heuristic log-return statistics pooled across all 24 runs. The clipped log return ($[-10, +10]$) is used throughout. Median is reported alongside mean as the primary location statistic; count indicates how many applications each heuristic received.

### 4.5 Footrule distances

Spearman footrule distance measures how far the ranking of heuristics by mean log return shifts between the early phase (first 25% of budget) and the late phase (last 25% of budget). Table 7

| Heuristic | Count | Mean | Median | IQR | Trim. mean | Note |
|---|---|---|---|---|---|---|
| 0 | 202,150 | +0.051 | 0.000 | 0.069 | +0.030 | |
| 1 | 201,309 | +0.014 | 0.000 | 0.013 | +0.005 | |
| 2 | 185,324 | +0.129 | 0.000 | 0.001 | +0.005 | |
| 3 | 177,048 | +0.085 | 0.000 | 0.034 | +0.013 | |
| 4 | 216,726 | +0.039 | 0.000 | 0.042 | +0.017 | |
| 5 | 202,359 | +0.059 | +0.003 | 0.091 | +0.040 | skew |
| 6 | 245,336 | +0.581 | 0.000 | 1.115 | +0.370 | skew |
| 7 | 280,936 | +0.373 | 0.000 | 0.000 | +0.059 | |
| 8 | 231,494 | +0.026 | 0.000 | 0.000 | +0.000 | |
| 9 | 184,278 | +0.098 | 0.000 | 0.000 | +0.000 | |
| 10 | 155,517 | +0.058 | 0.000 | 0.000 | +0.000 | |
| 11 | 93,240 | +0.763 | 0.000 | 1.962 | +0.605 | skew |
| 12 | 67,270 | +0.001 | 0.000 | 0.000 | +0.000 | |
| 13 | 28,484 | +0.000 | 0.000 | 0.000 | +0.000 | |
| 14 | 28,529 | +0.000 | 0.000 | 0.000 | +0.000 | |

Table 6: Per-heuristic log-return statistics pooled across all 25 runs and all six domains. Log returns clipped to $[-10, +10]$. Heuristic IDs are domain-relative indices; the same ID in different domains refers to different operators. Heuristics 5, 6, and 11 are flagged for distribution skew. Large mean with zero median indicates a long right tail on a small fraction of applications.

reports the distance per technique-domain run. The maximum attainable distance scales with the number of heuristics: a domain with $n$ heuristics has a maximum of $\lfloor n^2/2 \rfloor$.

| Domain | SP-CF-LA | SP-CF-AILTA | SP-SR-LA | MP-CF |
|---|---|---|---|---|
| BinPacking | 5 | 4 | 12 | 12 |
| FlowShop | 22 | 20 | 16 | 19 |
| PersonnelScheduling | 17 | 17 | 20 | 14 |
| SAT | 22 | 14 | 16 | 12 |
| TSP | 14 | 8 | 16 | 16 |
| VRP | 24 | 18 | 12 | 32 |
| Mean | 17.3 | 13.5 | 15.3 | 17.5 |

Table 7: Spearman footrule distances between early-phase (first 25%) and late-phase (last 25%) heuristic utility rankings. A distance of 0 means rankings are identical throughout; higher values mean greater rank instability. The maximum attainable distance depends on the number of heuristics per domain.

### 4.6 LIME condition frequency

LIME was applied at 50 evenly-spaced quantile positions across the budget for each per-technique surrogate. Condition records were produced for MP-CF; for the three single-point techniques the LIME grid produced empty condition sets. This is consistent with the class-imbalance problem in the single-point surrogates: when the acceptance rate is below 30%, LIME's local perturbation neighbourhood is dominated by rejection predictions and the local linear model has near-zero weights. Table 8 reports the top-5 conditions for MP-CF.

| Condition (MP-CF) | Count | Mean weight | Positive fraction |
|---|---|---|---|
| feat_gap_to_best $\leq 0.00$ | 25 | $-0.205$ | 0.00 |
| feat_obj_norm $\leq 0.00$ | 25 | $-0.114$ | 0.00 |
| feat_improve_rate_50 $\leq 0.00$ | 22 | $-0.029$ | 0.13 |
| feat_pop_target_rank $\leq 0.00$ | 22 | $+0.033$ | 1.00 |
| feat_pop_diversity $\leq 0.00$ | 21 | $-0.123$ | 0.00 |

Table 8: Top-5 LIME conditions by total frequency across the 50-point budget grid for MP-CF. Count is the number of grid points (out of 50) at which the condition appeared. Positive fraction is the proportion of those appearances where the condition pushed toward acceptance.

# 5 Analysis

## 5.1 Cross-domain performance patterns

Table 3 reveals that no single technique dominates across all six domains. SP-CF-AILTA wins on three domains (SAT, FlowShop, PersonnelScheduling), MP-CF wins on two (TSP and VRP), and the random baseline SP-SR-LA wins on one (BinPacking). SP-CF-LA, which pairs the Choice Function with the simpler acceptance criterion, does not lead on any domain and finishes last on TSP.

The AILTA advantage on SAT is large: 2.0 against 16.0 for the random baseline and 11.0 for SP-CF-LA. On FlowShop and PersonnelScheduling the margins are narrower (6297 vs 6346 and 3282 vs 3298 respectively). SAT minimises unsatisfied clauses toward a target of zero, so acceptance strategy matters greatly near the end of a run when the objective can be very close to zero. AILTA loosens its threshold when the search stagnates, which is precisely the condition needed to escape a local optimum with objective value 3 or 4.

The random baseline winning BinPacking is notable. BinPacking in HyFlex is a continuous-objective problem and the observed values are very close together (0.011–0.017). The differences may be within noise at a single seed; significance claims require replication. It does suggest that the Choice Function's learning is not helping on this domain at this instance and seed.

## 5.2 MP-CF budget fairness

MP-CF performs worst on SAT (49.0 vs 2.0 for AILTA) but best on TSP and VRP. Under an equal iteration budget of 100,000 a population of ten applies roughly 10,000 perturbations to each member, versus 100,000 for a single-point run. This is the budget-fairness problem identified in Assessment 3.

The TSP and VRP results suggest that the population approach provides a benefit *despite* the disadvantage, not because of it: population diversity helps in these domains more than the reduced per-member perturbation count hurts. On SAT the opposite is true: convergence to a near-zero objective requires sustained local search on one solution, and distributing the budget across ten members is inefficient.

A budget-scaled MP-CF run (using `budgetScaling: per_member`, which multiplies the budget by the population size before running) would apply 1,000,000 iterations total, making the comparison fairer in terms of perturbation count. This is available in the implementation but not executed in this assessment due to wall-clock cost; it is planned for the full matrix.

### 5.3 Surrogate failure for Simple Random reveals a CF learning effect

The most striking result in Table 4 is the SP-SR-LA ROC-AUC of 51.75%, which is statistically indistinguishable from random. SP-CF-LA, which uses the *same* Late Acceptance criterion, achieves 96.77% ROC-AUC on the same acceptance target. The only difference between the two techniques is the selection strategy.

The explanation is structural. Late Acceptance accepts a candidate if its objective is at least as good as the objective recorded $L = 50$ iterations ago. That historical comparison value is not a feature in the surrogate's input: the 16 state features capture recent trends (last 50 moves), stagnation counts, and heuristic history, but not the specific objective seen exactly $L$ steps ago. The surrogate must infer the LA threshold from proxies.

Under the Choice Function, selection is adaptive: productive heuristics are applied more often. This creates structure in the search trajectory that is visible in the features — sustained high `feat_improve_rate_50`, characteristic `feat_last_h1` patterns, correlations between quality features and subsequent acceptance — and the surrogate can exploit that structure. Under Simple Random, heuristics are applied uniformly at random regardless of recent outcome. The resulting trajectory has no such exploitable structure: the feature vector at decision time is not predictive of whether the candidate beats a historical objective value that the surrogate cannot observe.

This is the clearest evidence so far that the Choice Function does something observable beyond its role as a heuristic selector: its adaptive behaviour creates structured correlations in the search state that a surrogate can learn. The absence of that structure under Simple Random leaves the surrogate with near-zero discriminative power. Any explanation of the SP-SR-LA decisions based on a surrogate with 51.75% ROC-AUC would be explaining the surrogate's guesses rather than the actual policy. No SHAP attributions are reported for SP-SR-LA in this assessment.

### 5.4 Per-technique SHAP profiles

The key question for Assessment 4, following the pooling artefact identified in Assessment 3, is whether the Choice Function and Simple Random surrogates show different SHAP profiles for the non-population features. In the pooled surrogate population rank and diversity dominated precisely because they perfectly encoded topology membership. In the per-technique surrogates those features take sentinel value $-1$ for all single-point techniques, so the surrogate must explain acceptance on the basis of the remaining fourteen features.

Table 5 shows the full SHAP profiles. Several patterns are clear.

**feat_llh_count dominates the single-point surrogates.** This feature is the number of low-level heuristics available in the domain. Within a single run it is constant, but it differs across domains (SAT, BinPacking, FlowShop, TSP, VRP, and PersonnelScheduling all have different LLH counts). The per-technique surrogates are trained on pooled cross-domain decisions, so `feat_llh_count` acts as a domain identifier. The surrogate is learning domain-specific acceptance patterns via a domain label, not via domain-independent search-state features. This is the same kind of confound as the topology confound in Assessment 3, but at the domain level rather than the technique level. It is a direct consequence of pooling across domains.

**feat_iters_since_accept is critical for MP-CF but irrelevant for single-point.** For MP-CF, `feat_iters_since_accept` carries SHAP 86.23, the dominant feature by a large margin. For all three single-point techniques it is zero. In multi-point search, acceptance means population replacement. When no replacement has occurred for many iterations, the population has fully converged and the probability that the next candidate exceeds the worst population member rises sharply. This feature captures exactly that stagnation threshold. In single-point Late

Acceptance, the comparison is to the objective $L$ steps ago rather than to population state, so stagnation duration is less predictive.

**feat_gap_to_best is much more important with AILTA than with LA.** For SP-CF-AILTA, `feat_gap_to_best` has SHAP 10.86, second only to the domain identifier. For SP-CF-LA and SP-SR-LA it is 0.32 and 0.51. AILTA adapts its acceptance threshold based on recent improvement history. When the gap to best is small, AILTA may tighten the threshold; when it is large, AILTA loosens it. The surrogate correctly identifies this as the primary driver of AILTA's decisions.

**feat_improve_rate_50 is consistently important across all single-point techniques** (1.48 for SP-CF-LA, 1.31 for SP-CF-AILTA, 1.28 for SP-SR-LA) but much less so for MP-CF (0.11). Recent search momentum is the primary signal for single-point acceptance decisions regardless of selection strategy, while multi-point replacement depends more on stagnation duration and solution quality.

**The Choice Function creates more structured heuristic history signals.** The `feat_last_h1/h2/h3` features are more important in SP-CF-LA (0.39, 0.21, 0.43) than in SP-SR-LA (0.06, 0.03, 0.08). Under CF, specific heuristics are preferentially selected based on past performance, creating informative patterns in heuristic history that correlate with acceptance likelihood. Under Simple Random, heuristic history carries no signal.

**The expected AILTA stagnation signal does not appear.** `feat_iters_since_accept` has SHAP 0.000 for SP-CF-AILTA. This was not the predicted outcome: AILTA explicitly loosens its threshold when stagnation is detected. A possible explanation is that the stagnation effect is captured indirectly through `feat_iters_since_improve` (SHAP 0.317 for AILTA vs 0.155 for LA) and `feat_gap_to_best` (10.86 for AILTA). AILTA's adaptive loosening may manifest in solution-quality space rather than time-since-accept space.

### 5.5 Footrule distances: heuristic utility shifts across domains

Table 7 shows that heuristic utility rankings are not stable across the budget in any domain. The distances range from 4 (SP-CF-AILTA on BinPacking) to 32 (MP-CF on VRP), with most runs falling in the 12–24 range.

Two patterns stand out. First, SP-CF-AILTA has the lowest mean footrule distance (13.5) across all six domains, meaning its heuristic utility rankings are the most stable of the four techniques. This is not obviously expected: AILTA is an adaptive acceptance method that changes its threshold dynamically. One possible explanation is that AILTA's adaptive threshold reduces the noise in acceptance outcomes, leading to cleaner heuristic performance signals that are more consistent over time.

Second, MP-CF has the highest mean footrule distance (17.5) and achieves the single largest distance in the dataset: 32 for VRP. In multi-point search, the population maintains diversity and the fitness landscape seen by each member changes as the population evolves. Heuristics that help explore may be useful early and counterproductive late, or vice versa. This dynamic landscape creates more instability in heuristic rankings than the fixed incumbent in single-point search.

### 5.6 Log-return outliers and the median correction

Assessment 3 reported heuristic 6 at a mean log return of $+2.083$, roughly thirty times higher than the next best heuristic, and attributed this to near-zero-objective artefacts. Table 6 shows the corrected statistics with clipping to $[-10, +10]$ and with median alongside mean.

Heuristics 6 and 11 show the classic pattern: large positive mean, zero median, and very large IQR (1.115 and 1.962 respectively). The zero median indicates that the majority of applications of these heuristics produce no change, and the large mean is driven by a tail of high-return events. These heuristics are not systematically superior; they occasionally produce large improvements and otherwise do nothing. The trimmed mean (which removes the top and bottom 10% of values) is substantially lower than the raw mean for both heuristics (0.370 and 0.605 vs 0.581 and 0.763), further confirming the tail structure.

The correction directly addresses Assessment 3's item 6. Medians and counts are now reported, and skew diagnostics are printed and logged. A heuristic with mean 0.581 on 245,000 applications (heuristic 6) is qualitatively different from one with mean 0.058 on 155,000 applications (heuristic 10), and the new statistics make that difference visible.

### 5.7 Phase structure and the LIME grid

The 50-point LIME grid produced conditions only for MP-CF. The top five conditions (Table 8) are dominated by features taking value $\leq 0.00$, with `feat_gap_to_best` and `feat_obj_norm` both appearing at all 50 grid points and consistently pushing toward *rejection*. The condition `feat_pop_target_rank` $\leq 0.00$ appears at 22 points and consistently pushes toward *acceptance*: a candidate ranked at or below the median in the population strongly favours replacement.

The zero-threshold conditions reflect the behaviour of a multi-point surrogate with extreme class imbalance. When gap-to-best is zero, the candidate matches the best population member and replacement is unlikely; the LIME neighbourhood is also concentrated at zero because elitism protects the best. This is a LIME boundary artefact rather than a substantive finding.

For single-point techniques, the LIME grid produced no conditions. The next assessment should either (i) apply LIME to models trained with balanced class weights, or (ii) use LIME's alternative neighbourhood-construction for imbalanced targets. Direct measurement of acceptance rate against budget progress from the traces provides the model-free phase-structure evidence and does not suffer from this limitation.

### 5.8 Cross-domain generalisability

A core claim in the project is that the hyper-heuristic framework is domain-independent. The single-seed, single-instance results support this claim structurally: the same Java implementation ran unchanged across six substantially different problem types. The state features and surrogate architecture require no modification between domains.

What the results do not yet establish is whether the *explanation* is domain-independent. The SHAP profiles from the per-technique surrogates are trained on pooled cross-domain decisions; a domain-specific breakdown would require per-domain surrogates. This is the next logical step.

### 5.9 Limitations

Table 9 summarises the known limitations of this assessment and their consequences for interpreting the results.

| Limitation | Consequence |
|---|---|
| Single instance, single seed | No statistical claims can be made; results may reflect the difficulty profile of one instance. |
| Domain confound in per-technique surrogates | `feat_llh_count` acts as a domain label; SHAP attributions reflect domain identity as much as acceptance behaviour. |
| SP-SR-LA surrogate fails (ROC-AUC 51.75%) | No SHAP explanation is valid; Simple Random behaviour cannot be explained by this surrogate. |
| Unscaled MP-CF budget | MP-CF final-objective comparisons are structurally unfair; budget-scaled variant unrun. |
| LIME grid empty for single-point | Class imbalance prevents LIME from producing conditions for all three single-point techniques; phase structure hypothesis unvalidated. |
| No Wilcoxon tests | Technique ordering in Table 3 is preliminary; none is statistically significant. |
| Wall-clock dominated by PersonnelScheduling | Each PS run takes 5–8 hours; scaling to ten instances or five seeds requires approximately 50× the current runtime. |

Table 9: Limitations of the current assessment and their consequences.

# 6 Conclusion and Next Steps

Assessment 4 has delivered the six changes identified in Assessment 3. Real CHeSC HyFlex domains replace stubs, confirmed by `stub: false` in all 25 manifests. Per-technique surrogates are trained and compared. Log returns are stabilised with clipping, and medians and counts replace the sole-mean reporting. LIME is applied over a 50-point budget grid. The experimental matrix covers all six CHeSC domains.

The headline findings are as follows.

**Performance.** SP-CF-AILTA wins on three of six domains (SAT, FlowShop, PersonnelScheduling). MP-CF wins on two (TSP, VRP) despite a structural budget disadvantage. The random baseline wins on BinPacking.

**Surrogate faithfulness.** The SP-SR-LA surrogate achieves ROC-AUC 51.75%, indistinguishable from random. This is a new finding: without the Choice Function's adaptive selection, the 16 state features do not predict Late Acceptance decisions. The CF creates structured correlations in search state that the surrogate can learn; Simple Random does not.

**SHAP profiles.** The per-technique surrogates reveal that `feat_llh_count` (LLH count, constant within a run) dominates single-point surrogates trained on pooled cross-domain data, indicating a domain-identity confound analogous to the topology confound in Assessment 3. Excluding this artefact, `feat_improve_rate_50` is the primary signal for single-point techniques while `feat_iters_since_accept` and `feat_iters_since_improve` dominate MP-CF. `feat_gap_to_best` is substantially more important for AILTA than for LA, which is consistent with AILTA's quality-sensitive threshold.

**Work for the next submission:**

1. Train per-domain surrogates (excluding `feat_llh_count`) to remove the domain-identity confound and obtain domain-independent SHAP profiles.

2. Resolve the LIME imbalance problem for single-point techniques using balanced class weights or targeted sampling.
3. Run budget-scaled MP-CF to make multi-point final-objective comparisons fair.
4. Run the full matrix (four techniques, six domains, ten instances, five seeds) and apply Wilcoxon signed-rank tests with Holm correction.
5. Sweep Late Acceptance list length over $\{1, 5, 50, 100, 500, 1000\}$ and population size over $\{5, 10, 20, 50\}$.

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